Representation as Evidence: Representation Aware Deep Evidential Regression

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Abstract

Deep Evidential Regression (DER) estimates aleatoric and epistemic uncertainty through a Normal-Inverse-Gamma (NIG) predictive distribution in a single forward pass. However, evidential parameters can become unreliable under distribution shift, particularly in high-dimensional visual regression, where weakly supported latent regions may still receive confident evidence. We introduce Representation-Aware DER, a support-guided evidential framework that explicitly couples latent representation support with evidential uncertainty. The proposed method estimates internal latent support using a memory bank of normalized representations and uses it to condition the parameterization of the NIG virtual evidence parameter v, while preserving the original DER objective and NIG output family. As a result, weakly supported regions receive reduced effective evidence, leading to increased epistemic uncertainty under distribution shift. We further adopt a progressive training strategy to facilitate the optimization of representation-aware evidential learning. We evaluate the method on synthetic regression and monocular depth estimation using NYU Depth as in-distribution data and SUN RGB-D, ApolloScape, and Cityscapes under distribution shift. Experiments across multiple evidential signals, collapse indicators, and post-hoc latent-space analyses demonstrate improved ID/OOD uncertainty separation over conventional DER and competitive performance against other uncertainty baselines, while highlighting the remaining sensitivity to training dynamics. These results suggest that latent support provides a practical mechanism for improving evidential uncertainty reliability without requiring OOD supervision, Bayesian sampling, or ensemble inference.

1. Introduction

Since the introduction of DER by Amini et al. [1], evidential learning has received growing attention as an efficient alternative to Bayesian sampling and ensemble-based uncertainty estimation. Subsequent studies explored evidential uncertainty across regression, classification, medical imaging, autonomous driving, and dense prediction tasks, motivated by the ability of evidential models to estimate predictive uncertainty in a single forward pass. Despite its empirical success, recent work has raised important theoretical and practical concerns regarding evidential regression. Meinert et al. [10] argue that conventional DER behaves more as a heuristic uncertainty estimator than a fully grounded Bayesian uncertainty framework, highlighting ambiguities in the interpretation of evidential parameters and epistemic uncertainty extraction. Wu et al. [19] further identify the evidence contraction issue, showing that the non-negativity constraints imposed on NIG parameters can produce suboptimal optimization behavior and unstable evidence dynamics. More recently, Chen et al. [2] revisit the essential and nonessential components of evidential deep learning, while Pandey et al. [12] analyze unstable evidence accumulation and zero-evidence learning regions under evidential optimization. In parallel, representation-aware uncertainty estimation methods have shown that the geometry of learned latent representations contains important information about distribution shift and model reliability. Latent-space density estimation, Mahalanobis-distance scoring [8], and deep nearest-neighbor methods [16] have demonstrated strong effectiveness for out-of-distribution (OOD) detection by identifying samples that deviate from the training representation manifold. These approaches suggest that uncertainty should depend not only on model predictions, but also on the degree of support within the learned feature space. However, most existing representation-based OOD approaches operate as post-hoc scoring mechanisms separate from evidential uncertainty estimation, while evidential regression methods typically estimate uncertainty directly from predictive evidence without explicitly modeling latent representation support. Related probabilistic approaches also connect predictive uncertainty to representation support or distance. Evidential Conditional Neural Processes (ECNP) and Distance-informed Neural Processes (DNP) incorporate context–target relationships or distance information into uncertainty estimation, while GP-based approaches such as GroVE perform post-hoc proba-

bilistic uncertainty estimation in learned embedding spaces [11, 17, 18]. These methods reinforce the importance of representation support, but differ from our setting: ECNP and DNP rely on task-provided context sets, while GP-based embedding methods operate as post-hoc uncertainty estimators. In contrast, our method uses a periodically refreshed memory bank containing a subset of training representations to estimate latent support and condition the NIG virtual-evidence parameter during optimization. Thus, representation support directly affects evidential uncertainty generation rather than serving as a separate post-hoc score or calibration layer. Intuitively, samples located in weakly supported latent regions should not receive the same evidential confidence as samples that closely match the training distribution. If virtual evidence could adapt according to latent representation support, unsupported regions could naturally induce evidence collapse and stronger epistemic uncertainty under OOD shift. Recent work has shown that evidential models can suffer from unstable evidence dynamics and zero-evidence learning regions [12], while OOD detection methods have explored energy-based formulations for identifying unsupported inputs [9, 20]. These observations suggest that evidential strength is closely related to latent representation support. High-dimensional regression tasks in computer vision, particularly monocular depth estimation, provide challenging benchmarks for uncertainty estimation under realistic distribution shift and dense prediction settings [6, 13]. Existing uncertainty methods for dense visual regression typically rely on predictive variance, Bayesian sampling, or ensemble disagreement without explicitly coupling uncertainty estimation to latent representation support. In this work, we introduce Representation-Aware DER, a framework that explicitly couples latent-space structure with evidential uncertainty estimation. Instead of modifying the original DER loss or introducing auxiliary OOD supervision, we preserve the original NIG formulation while conditioning the virtual evidence parameter υ on local latent-space support estimated from representation similarity. The resulting support score modulates evidential strength, increasing epistemic uncertainty in weakly supported regions. The proposed formulation introduces several desirable properties. First, uncertainty estimation becomes representation-aware rather than depending solely on prediction residuals. Second, weakly supported OOD inputs can induce reduced effective evidence without requiring explicit OOD labels, Bayesian sampling, or ensemble inference. Third, the framework preserves single-model inference while incorporating latent representation support into evidential uncertainty estimation. Finally, the proposed method directly couples latent-space geometry with uncertainty generation, allowing predictive uncertainty and representation structure to interact within a unified evidential framework rather than operating as separate OOD detection modules. We evaluate the proposed framework on both synthetic regression and high-dimensional monocular depth estimation under distribution shift. Experiments on NYU Depth and ApolloScape demonstrate improved ID–OOD uncertainty separation compared with conventional DER. In particular, evidence-collapse indicators and representation-aware epistemic uncertainty provide strong OOD discrimination under distribution shift. These results suggest that conditioning evidential strength on latent-space support can improve uncertainty estimation under distribution shift. Our main contributions are summarized as follows: **We introduce Representation-Aware DER**, which conditions the NIG virtual evidence parameter υ on latent-space support to improve epistemic uncertainty estimation under distribution shift. **We introduce a progressive evidential training strategy** to improve the optimization of our representation-aware evidential model, using an initial predictive-training stage before incorporating latent support into evidential learning. **We analyze multiple evidential uncertainty signals and evidence-collapse** indicators under distribution shift, including predictive, epistemic, aleatoric, and collapse-based uncertainty formulations, showing that collapse-aware evidential signals provide improved ID–OOD discrimination.

2. Method

The proposed framework introduces latent-support-guided virtual evidence estimation for DER. Rather than estimating evidential strength solely from prediction residuals and network activations, the proposed method guides the evidential head using latent-space support information. Specifically, local support within the learned representation manifold is estimated using a k-nearest-neighbor memory bank and injected into the NIG virtual evidence parameter υ. This coupling encourages reduced evidential strength in weakly supported latent regions while preserving the original DER formulation. Finally, we introduce a progressive evidential training strategy that first allows the latent representation manifold to organize before progressively enabling support-guided uncertainty learning.

2.1. Representation-Aware Evidential Regression

DER models predictive uncertainty using a Normal-Inverse-Gamma (NIG) distribution parameterized by $(\gamma, \upsilon, \alpha, \beta)$, where γ denotes the predictive mean and υ, α, β control evidential uncertainty. The standard DER [1] objective is:

$$\mathcal{L}_{DER} = \mathcal{L}_{NLL} + \lambda \mathcal{L}_{reg}, \tag{1}$$

where $\mathcal{L}_{NLL}$ is the NIG negative log-likelihood and $\mathcal{L}_{reg}$ penalizes confident incorrect predictions. Existing DER methods learn evidential parameters directly from prediction residuals and network activations through the NIG parameterization [1]. Although DER can produce meaningful

uncertainty behavior in practice, prior studies have shown that evidential parameters may become unstable under distribution shift [10, 19]. In particular, the virtual evidence parameter υ, which controls epistemic uncertainty in DER, may remain relatively large for weakly supported latent representations. To address this limitation, we introduce a support-guided evidential framework that modulates virtual evidence using latent support.

Training Objective: Proposed method preserves the original DER training objective introduced in Eq. 1. Rather than modifying the optimization objective, the proposed framework injects latent-support information directly into the virtual evidence parameterization through υ_{eff}. Consequently, the method preserves the original DER formulation.

2.2. Latent-Space Support Estimation

Given an input sample x, the encoder bottleneck produces a latent feature map. Global average pooling over the spatial dimensions is applied to obtain a compact latent representation $z = \phi_\theta(x)$, followed by ℓ_2 normalization. Normalization is used because cosine similarity is implemented as a dot product between normalized representations; thus $d(\bar{z}, \bar{z}_i) = 1 - \bar{z}^\top \bar{z}_i$ gives the cosine distance and prevents feature magnitude from affecting support. A latent memory bank $\mathcal{B} = \{\bar{z}_1, \ldots, \bar{z}_M\}$ is constructed from normalized training representations and periodically refreshed during training to track the evolving latent manifold. For a query representation, cosine distances to the memory bank are computed and the K nearest latent neighbors are selected. The proposed framework estimates latent support from the local neighborhood structure of the learned representation manifold using a Gaussian kernel over nearby latent samples:

$$S(x) = \frac{1}{K} \sum_{i \in \mathcal{N}_K(x)} \exp\left(- \left(\frac{d_i}{h\tau + \epsilon} \right)^2 \right),$$

where d_i denotes cosine distance, h controls the kernel bandwidth, and τ is a temperature parameter controlling support smoothness. For the adaptive-bandwidth variant, h is estimated from the support bank: for each bank feature, we compute its nearest-neighbor distance to all other bank features, excluding the self-match, and set h to the 90th percentile of these distances. This provides a data-dependent kernel scale for the Gaussian support kernel. To prevent numerical collapse, support is lower bounded as:

$$\hat{S}(x) = S_{\min} + (1 - S_{\min}) S(x), \tag{2}$$

where $S_{\min}$ defines the minimum allowed latent support. Support estimation is performed in the normalized latent representation space, allowing the evidential uncertainty structure to adapt according to local representation density. The resulting support score reflects the local support density of a sample within the learned latent manifold. Intuitively, samples located in densely populated latent neighborhoods receive stronger support estimates, while isolated or weakly supported regions naturally reduce effective virtual evidence and influence downstream uncertainty behavior. To improve robustness, we additionally evaluate an adaptive bandwidth variant in which the kernel scale is estimated from nearest-neighbor statistics within the support bank. This refinement dynamically adjusts the support scale according to the local latent representation density while preserving the original support-guided evidential formulation. Although maintaining a global memory bank can be computationally expensive, we utilize compact latent representations obtained through Global Average Pooling. A memory bank size of (M=512) was empirically sufficient to represent the global latent manifold for depth-estimation tasks without incurring substantial $(O(N \cdot M))$ search overhead. To track the evolving representation manifold during optimization, the support bank is periodically rebuilt every 1000 iterations, ensuring that support estimates remain aligned with the current feature space.

2.3. Support-guided Evidential Head

The network predicts the raw NIG parameters $(\gamma, \upsilon_{\text{raw}}, \alpha, \beta)$. Unlike conventional DER, the proposed framework applies support guidance to the virtual evidence parameter using latent-space support statistics. Since epistemic uncertainty in DER is inversely proportional to υ, support-guided modulation directly couples evidential strength with latent representation density.

$$\upsilon_{\text{eff}}(x) = \hat{S}(x)\, \upsilon_{\text{raw}}(x). \tag{3}$$

The remaining parameters are left unchanged, yielding the final NIG distribution $(\gamma, \upsilon_{\text{eff}}, \alpha, \beta)$. This formulation introduces an explicit dependency between epistemic uncertainty and latent-space support. Weakly supported latent regions therefore induce reduced virtual evidence and increased epistemic uncertainty. Importantly, the proposed method preserves the original DER optimization objective and NIG evidential output family while modifying the parameterization of virtual evidence through latent-support guidance, without requiring auxiliary supervision, Bayesian sampling, or ensemble training. Algorithm 1 summarizes the proposed support-guided evidential training pipeline. The support bank is periodically refreshed using training representations, while support guidance is applied only to the virtual observation parameter υ.

2.4. Uncertainty Estimation

We analyze evidential degradation under distribution shift using our collapse-based signals: $-\log(\upsilon_{\text{eff}})$, $-\log(\alpha - 1)$, $-\log(\beta)$, and $-\log(\upsilon_{\text{eff}}(\alpha - 1))$, which respectively

Algorithm 1 Support-guided Evidential Regression

Require: $D = \{(x_i, y_i)\}_{i=1}^{N}$, encoder ϕ_θ, regressor f_θ, neighbors K, bank size M, support floor $S_{\min}$, bandwidth h, temperature τ, warm-up T_{prog}; Initialize θ and support bank $\mathcal{B}$

for each training iteration t **do**
  Sample minibatch (x, y)
  if support bank update step **then**
    Rebuild $\mathcal{B}$:z_i=Normalize(Pool($\phi_\theta(x_i)$)), i=1...M
  Compute $z = \text{Normalize}(\text{Pool}(\phi_\theta(x)))$
  Select K nearest neighbors of z in $\mathcal{B}$ by cosine distance
  Estimate latent-space support:
  $\hat{S}(x) = S_{\min} + (1 - S_{\min}) \frac{1}{K} \sum_{i \in \mathcal{N}_K(x)} \exp\left(- \left(\frac{d_i}{h\tau + \epsilon} \right)^2 \right)$
  if $t \leq T_{\text{prog}}$ **then**
    Compute prediction objective: $\mathcal{L} = \|y - \gamma\|_2^2$
  else
    Predict raw NIG parameters: $(\gamma, \upsilon_{\text{raw}}, \alpha, \beta) = f_\theta(x)$
    Apply support guidance: $\upsilon_{\text{eff}} = \hat{S}(x)\upsilon_{\text{raw}}$
    Compute DER objective: $\mathcal{L} = \mathcal{L}_{NLL} + \lambda \mathcal{L}_{reg}$
  Update network parameters θ

capture virtual-evidence degradation, evidential-confidence degradation, scale degradation, and joint evidential collapse. For completeness, we also report the standard DER aleatoric and epistemic uncertainty estimates:

$$u_{\text{alea}} = \frac{\beta}{\alpha - 1}, \qquad u_{\text{epis}} = \frac{\beta}{\upsilon_{\text{eff}}(\alpha - 1)}. \tag{4}$$

Progressive Representation-Aware Evidential Training. The proposed framework uses latent-space support statistics to guide evidential uncertainty estimation. Consequently, the quality of the latent representation manifold directly influences the resulting evidential behavior. In preliminary experiments, directly enabling support-guided evidential optimization from random initialization frequently produced unstable uncertainty dynamics and poor ID/OOD separation. We hypothesize that this occurs because the latent feature space has not yet formed sufficiently meaningful neighborhood structure during the early stages of optimization. To address this issue, we adopt a progressive training strategy. The model is first optimized using standard predictive supervision before enabling support-guided evidential optimization. This staged optimization allows the latent representation manifold to partially organize before support-guided evidential uncertainty learning begins.

3. Experiments

We evaluate the proposed support-guided evidential framework on both controlled synthetic regression and monocular depth estimation under distribution shift. The toy regression experiments analyze uncertainty behavior and support-guided evidential dynamics under noisy extrapolation settings. The depth-estimation experiments further evaluate uncertainty quality, representation-space organization, post-hoc kNN latent-space separation, and optimization stability under realistic near- and far-OOD visual domain shifts. All depth-estimation methods use the same U-Net-style encoder–decoder backbone [14] and are optimized using Adam with a learning rate of 5×10^{-5}, gradient clipping norm 1.0, and batch size 4. Evidential methods are trained for 25,000 iterations in total; progressive variants use the first 5,000 iterations for predictive training, followed by 20,000 iterations of evidential optimization. The proposed support-estimation module uses a memory bank size of M=512, K=50 nearest neighbors, kernel temperature τ=8.0, and minimum support floor $S_{\min}$=.05. To track the evolving latent manifold during optimization, the support bank is rebuilt every 1,000 iterations. Unless otherwise stated, methods use identical backbone architectures, optimization settings, and checkpoint-selection criteria to ensure fair comparison across uncertainty formulations. OOD evaluation is performed on SUN RGB-D as near-OOD, ApolloScape as a substantial domain shift, and Cityscapes as a stronger far-OOD benchmark. Results are aggregated across independent runs as specified in each table. For each run, the checkpoint with the best validation RMSE is used for OOD evaluation. To assess training stability, we independently repeated the complete training procedure using the same configuration and fixed random seed. Despite these controlled settings, we observed variability in optimization outcomes and OOD uncertainty estimates across runs.

3.1. Training Strategy and Stability

For evidential methods, we evaluate both direct scratch optimization and progressive training. In scratch training, support-guided evidential optimization is enabled immediately from random initialization. In progressive training, the model is first optimized using standard predictive supervision before support-guided evidential learning is enabled. This staged optimization allows the latent representation manifold to partially organize before uncertainty estimation begins. In preliminary experiments, directly enabling support-guided evidential optimization from initialization frequently produced unstable uncertainty behavior and substantial run-to-run variability. Although some runs achieved meaningful ID/OOD separation, uncertainty behavior varied substantially across checkpoints and independent runs. In unstable cases, latent support degenerated early during optimization, causing evidential quantities to become poorly separated under distribution shift. Across near-OOD and far-OOD evaluations, progressive training improves mean OOD discrimination for selected uncertainty signals, although individual scratch runs can achieve stronger AUROC and run-to-run variability remains substantial. Across 10 attempted progressive-training runs, one entered a catas-

trophic degenerate state with collapsed or NaN-valued uncertainty responses and near-random ID/OOD discrimination ($\mathrm{AUROC} \approx 0.5$). This run was rejected by the numerical-validity criterion, and quantitative results therefore report the remaining nine runs. We define such degenerate runs by collapsed uncertainty behavior, NaN-valued responses, or near-random OOD discrimination, and analyze them separately from numerically valid runs. Importantly, these degenerate states are explicit and readily detectable through near-random uncertainty separation, collapsed evidential responses, and failure of uncertainty-map generation. This suggests that coupling evidential strength with latent representation support allows instability in the learned support manifold to appear as observable uncertainty degradation rather than silently producing misleading confidence estimates. We also observe a correlation between the final optimization objective and uncertainty reliability. Stable runs consistently reach a lower training loss floor, approximately -2.0 in our depth experiments, whereas non-rejected failure cases reach a higher loss floor, approximately -1.5. This provides a useful diagnostic indicator of sub-optimal ID/OOD uncertainty separation. We evaluated support-guided joint conditioning of (υ, α). Empirically, conditioning the virtual evidence parameter υ alone produced the most stable uncertainty behavior and strongest ID/OOD separation across runs. We hypothesize that this occurs because epistemic uncertainty in DER is directly inversely proportional to υ, allowing support modulation to influence epistemic uncertainty without substantially perturbing the remaining evidential variance structure.

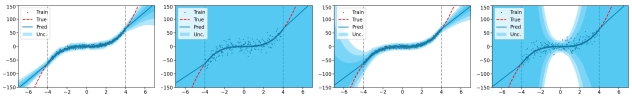
3.2. Compared Methods

We compare the proposed support-guided evidential framework against multiple uncertainty estimation and evidential regression baselines under identical depth-estimation and OOD evaluation settings. We compare against conventional DER [1], which models predictive uncertainty using a Normal-Inverse-Gamma (NIG) evidential distribution learned directly from regression supervision. We additionally compare against the recent evidential regression formulation proposed in [19], which introduces an alternative evidential optimization objective for uncertainty-aware regression. We evaluate Deep Ensembles [7], where uncertainty is estimated from disagreement across independently trained models. Ensemble predictive variance and predictive standard deviation are used as the primary OOD uncertainty signals. We evaluate Monte-Carlo Dropout [4], where uncertainty is estimated from repeated stochastic forward passes during inference. For post-hoc representation-space analysis, we additionally evaluate Mahalanobis distance-based OOD scoring [8] computed in the penultimate latent feature space, as well as k-nearest-neighbor representation scoring [16]. All methods are evaluated on monocular depth estimation using the NYU Depth V2 dataset [15]. All depth-estimation methods use the same encoder–decoder backbone and training protocol, with only the uncertainty-estimation mechanism changed across methods. Uncertainty estimation is performed directly on the depth regression output using evidential, ensemble-based, or sampling-based uncertainty formulations depending on the evaluated method. We use ApolloScape [5] as the primary near-OOD benchmark due to its substantial domain shift relative to indoor NYU Depth scenes while still preserving realistic depth structure and driving-scene geometry. To further evaluate robustness under stronger distribution shift, we additionally evaluate on Cityscapes [3] as a far-OOD benchmark. For Cityscapes, disparity maps are resized and normalized following standard preprocessing used in monocular depth-estimation pipelines. Both datasets are evaluated without additional retraining or domain adaptation. To ensure experimental fairness, we additionally evaluate progressive and scratch-training variants of both the original DER [1] and the recent evidential formulation proposed in [19] using the same progressive-training strategy and identical backbone architecture as the proposed method. The proposed support-guided method achieves higher mean Collapse-β AUROC than conventional DER across all three OOD datasets. However, the gains over other evidential methods are mixed, and run-to-run variability remains substantial.

3.3. Synthetic Regression Under Distribution Shift

We first evaluate the proposed support-guided evidential regression framework on a synthetic cubic regression task commonly used in DER evaluation. The training data are sampled from $x \in [-4, 4]$, while testing is performed on the wider interval $x \in [-7, 7]$ to evaluate extrapolation behavior and uncertainty calibration under distribution shift. Following prior DER work, the target function is defined as $y = x^3 + \epsilon$. We consider two training settings: **Low-noise setting:** homoscedastic Gaussian noise with σ=3. **High-noise corrupted setting:** 20% of training samples are corrupted using large-variance noise (σ=15), while the remaining samples use $\sigma = 1$. The proposed model uses a two-layer MLP backbone with a latent support bank constructed from normalized latent representations. Support is estimated using cosine similarity over the local latent neighborhood. The support score modulates only the NIG virtual observation parameter υ. We compare against the original DER formulation of Amini et al., implemented using the same backbone architecture and training objective for fair comparison. Both methods are trained using the evidential NIG loss with identical optimization settings.

Behavior Under High Noise Figure 1 shows the predictive uncertainty under severe training corruption. The original DER formulation becomes overconfident in out-of-



a. & b. Amini [1] c. & d. Ours

Figure 1. Regression under Gaussian and corrupted Gaussian-mixture noise. DER becomes overconfident under severe corruption, whereas the proposed support-guided DER produces stronger uncertainty expansion outside supported latent regions.

Table 1. ApolloScape OOD evaluation across nine runs. Results are reported as mean ± standard deviation. All metrics use the OOD-separating polarity of each signal. C. denotes Collapse.

	Ours Scratch			Ours Progressive		
Signal	AUROC↑	AUPR↑	FPR95↓	AUROC↑	AUPR↑	FPR95↓
Std.	.770±.12	.712±.14	.523±.17	.809±.12	.761±.14	.493±.20
Epistemic	.742±.09	.676±.11	.591±.15	.784±.10	.727±.12	.564±.18
Aleatoric	.760±.10	.696±.13	.553±.15	.778±.10	.719±.11	.593±.18
C.-υ	.698±.14	.640±.15	.597±.17	.732±.18	.681±.19	.532±.28
C.-α	.748±.15	.685±.18	.506±.17	.759±.16	.714±.18	.526±.28
C.-β	.778±.15	.729±.17	.488±.19	**.814±.15**	**.774±.17**	**.444±.23**
C.-Joint	.744±.12	.680±.16	.547±.15	.728±.17	.673±.18	.531±.28
Evidence	.701±.16	.650±.17	.583±.19	.762±.17	.717±.19	.540±.27

Table 2. ApolloScape OOD evaluation across nine runs. Evidential signals include Collapse-$\beta = -\log(\beta)$, and the support score $S = 1/\hat{S}(x)$, where $\hat{S}(x)$ denotes latent support. Mean μ and variance σ^2 are reported.

	Amini Scratch [1]				**L2024 progressive [19]**				**Ours Scratch**				**Ours progressive**				
	Std.	**Epis.**	**Alea.**	β	**Std.**	**Epis.**	**Alea.**	β	**Std.**	**Epis.**	**Alea.**	β	**Std.**	**Epis.**	**Alea.**	β	S
Mean	.772	.756	.743	.765	.796	.764	.775	.817	.770	.742	.760	.778	**.809**	**.784**	**.778**	**.814**	.679
Var	.008	.006	.006	.013	.028	.019	.019	.031	.014	.009	.010	.022	.014	.011	.009	.022	.008

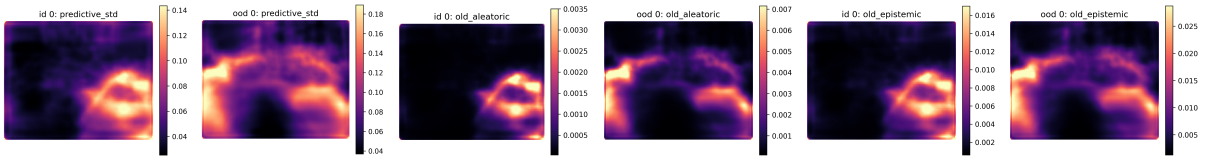
distribution regions and produces unstable uncertainty estimates under heavy noise corruption. In contrast, the proposed support-guided evidential model increases predictive uncertainty outside the supported latent manifold while maintaining stable in-distribution predictions. Additional support-estimation visualizations and the low-noise sanity check are provided in the appendix. These results confirm that latent support decreases away from densely represented regions. As a sanity check, we additionally evaluate the standard low-noise cubic setting. Under clean training conditions, both methods produce stable predictions and reasonable extrapolation uncertainty, indicating that support guidance does not degrade conventional DER behavior. The advantage of the proposed method becomes substantially more pronounced under corrupted high-noise training, where standard DER frequently exhibits unstable or overconfident uncertainty estimates.

3.4. Uncertainty Metrics

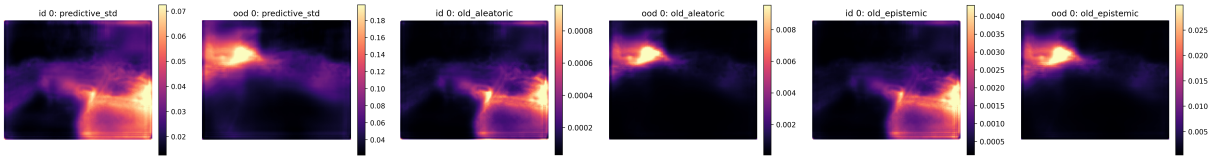
Beyond predictive standard deviation, we evaluate several evidence-collapse signals derived from the NIG parameters, including Collapse-υ, Collapse-α, Collapse-β, and joint collapse. Tables 1 and 2 report quantitative OOD performance across the evaluated uncertainty signals, while Figures 2 and 3 visualize their corresponding uncertainty behavior under distribution shift. Among the evaluated signals, Collapse-β and predictive standard deviation frequently provide the clearest qualitative ID/OOD separation across independent runs. The qualitative visualizations shown in Figures 2 and 3 correspond to Run 1, a representative stable progressive-training run exhibiting strong uncertainty separation across predictive and collapse-based signals. For this reason, we analyze both ranking-based metrics and qualitative uncertainty structure throughout the experiments.

3.5. Depth Estimation Under Distribution Shift

We evaluate the proposed representation-aware evidential regression framework on monocular depth estimation under distribution shift. Models are trained on NYU Depth V2 and evaluated on SUN RGB-D as near-OOD, ApolloScape as a substantial domain shift, and Cityscapes as a stronger far-OOD benchmark. All methods use the same encoder–decoder backbone, optimizer, training schedule, and checkpoint-selection criterion. The primary difference lies in the evidential parameterization strategy: conventional DER predicts evidential precision υ directly from network outputs, whereas the proposed method conditions evidential strength on latent representation support estimated from the learned feature manifold. Following prior evidential regression literature, uncertainty quality is evaluated using AUROC, AUPR, and FPR95 together with predictive, epistemic, aleatoric, and evidence-collapse uncertainty measures. Table 3 reports OOD detection performance. Conventional DER shows substantial ID/OOD uncertainty overlap, whereas support-guided DER improves discrimination by reducing evidential precision in weakly supported latent regions. Collapse-based signals, particularly Collapse-β, provide the strongest separation across the evaluated distribution shifts. Figure 2 compares representative uncertainty maps from conventional DER and the proposed method under distribution shift. However, because latent support is estimated from globally pooled representations, these visualizations do not establish improved pixel-level uncertainty localization. Figure 3 further compares score distributions for NYU Depth and ApolloScape. Conventional DER exhibits substantial overlap between ID and OOD predictive and epistemic uncertainty distributions, indicating unreliable uncertainty separation. In contrast, the proposed support-guided evidential formulation produces substantially cleaner separation across multiple evidential signals, particularly for collapse-based metrics. Although scratch optimization occasionally achieves higher AUROC on individual uncertainty signals, progressive training improves mean Collapse-β AUROC and FPR95 across all three evaluated OOD datasets. However, run-to-run variability remains substantial, and improvements vary across

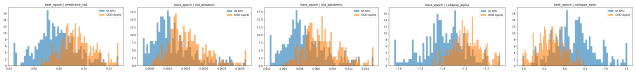


Amini DER [1] (scratch): Predictive std., aleatoric, and epistemic (ID/OOD)

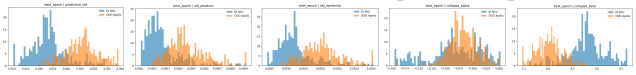


Ours (progressive): Predictive std., aleatoric, and epistemic (ID/OOD)

Figure 2. Qualitative uncertainty maps for NYU Depth (ID) and ApolloScape (OOD), comparing conventional DER [1] with our proposed support-guided method.



Baseline scratch [1]: std, Aleatoric, Epistemic, alpha Beta



Ours Progressive training: std, Aleatoric, Epistemic, alpha Beta

Figure 3. OOD score distributions for NYU (ID) and ApolloScape(OOD) across [1] and ours. Ours produces stronger ID/OOD separation and more structured uncertainty behavior.

Table 3. OOD detection performance on ApolloScape, Cityscapes, and SUN RGB-D. Evidential methods use Collapse-$\beta = -\log(\beta)$, while MC Dropout and Deep Ensembles use predictive standard deviation. Results report the mean (μ) and variance (σ^2) of AUROC and FPR95 across nine runs. S and P denote scratch and progressive training, respectively.

Method	**Apollo(Near-OOD)**				**City(Far-OOD)**				**SUN(Near-OOD)**			
	AUROC↑		**FPR95↓**		**AUROC↑**		**FPR95↓**		**AUROC↑**		**FPR95↓**	
	μ	σ^2	μ	σ^2	μ	σ^2	μ	σ^2	μ	σ^2	μ	σ^2
Amini-S	.765	.0133	.542	.0143	.786	.0273	.414	.0560	.703	.0230	.722	.0346
Amini-P	.719	.0279	.573	.0364	.830	.0116	.388	.0187	.679	.0220	.768	.0080
L2024-S	.773	.0173	.585	.028	.871	.0045	.380	.0179	.758	.004	.761	.003
L2024-P	**.817**	.0310	.426	.0605	.880	.0147	.274	.0421	.785	.023	.686	.021
Dropout	.767	.0146	.557	.0152	.810	.0126	.432	.0223	.744	.0057	.745	.0026
Ensemble	**.866**	.0265	**.289**	.1010	.861	.0378	.233	.0888	**.842**	.0317	**.366**	.1033
Ours-S	.778	.0220	.488	.0369	.883	.0097	.289	.0390	.751	.0124	.708	.0140
Ours-P	.814	.0217	.444	.0531	**.905**	.0160	**.193**	**.0293**	**.799**	.0172	**.637**	.0386

uncertainty signals.

3.6. KNN Representation Analysis

To analyze the learned representation geometry, we perform post-hoc OOD evaluation using k-nearest-neighbor (KNN) distance statistics [16] and Mahalanobis scoring [8] in the penultimate feature space. We use $k = 50$ as the representative KNN setting. Table 4 summarizes latent-space OOD separation across methods using both local neighborhood statistics and global covariance-based representation scoring. Deep Ensemble achieves the highest mean AUROC across all evaluated latent-space metrics, with the lowest run-to-run variability. Nevertheless, the proposed progressive method achieves competitive latent-space separation on far-OOD Cityscapes. Compared with scratch training, progressive training improves mean KNN AUROC from 0.878 to 0.943 and the 50th-neighbor KNN AUROC from 0.856 to 0.935, while reducing their standard deviations from 0.109 to 0.044 and from 0.116 to 0.051, respectively. Mahalanobis AUROC remains approximately unchanged at 0.919. On ApolloScape, scratch training achieves slightly higher mean latent-space AUROC than progressive training across the three distance metrics. Importantly, post-hoc latent-space separation alone is not sufficient to guarantee reliable evidential uncertainty. While Table 4 shows mixed results for representation-space distances, the uncertainty-based results show improved mean OOD discrimination for the proposed progressive method using Collapse-β across ApolloScape, Cityscapes, and SUN RGB-D. This distinction highlights the support–prediction gap addressed by the

Table 4. Post-hoc latent-space OOD detection using kNN ($k = 50$) and Mahalanobis distances on ApolloScape (Apollo) and Cityscapes (City), with NYU Depth V2 as the in-distribution dataset. Results report AUROC mean ± standard deviation across nine runs (higher is better). The highest mean AUROC in each column is bold.

Method		kNN-50 mean distance		kNN-50 kth distance		Mahalanobis distance	
Progressive		Apollo	City	Apollo	City	Apollo	City
Amini	✗	.732 ± .106	.826 ± .130	.703 ± .125	.820 ± .139	.778 ± .074	.879 ± .106
Amini	✓	.793 ± .086	.922 ± .034	.769 ± .098	.933 ± .031	.798 ± .050	.910 ± .059
Loss2024	✗	.783 ± .118	.844 ± .095	.777 ± .117	.861 ± .104	.771 ± .098	.842 ± .090
Loss2024	✓	.758 ± .054	.911 ± .072	.723 ± .062	.912 ± .067	.749 ± .053	.917 ± .048
Dropout	✗	.608 ± .053	.763 ± .159	.557 ± .063	.767 ± .177	.728 ± .050	.800 ± .110
Ensemble	✗	**.873 ± .025**	**.969 ± .019**	.858 ± .025	.967 ± .017	**.855 ± .020**	**.952 ± .016**
Ours	✗	.825 ± .091	.878 ± .109	.793 ± .108	.856 ± .116	.823 ± .084	.919 ± .072
Ours	✓	.812 ± .092	.943 ± .044	.769 ± .113	.935 ± .051	.790 ± .069	.919 ± .068

Table 5. AUROC μ, σ^2 over nine runs. W; Warmup. S: support-only. PS: post-hoc fusion with support.

Dataset	Amini+S	Amini+PS	Amini-W+S	Amini-W+PS	Ours
Apollo	.657 .017	.745 .036	.706 .011	.799 .013	**.814 .022**
SUN	.541 .003	.640 .019	.570 .001	.666 .005	**.799 .017**
City	.796 .022	.838 .026	.840 .005	.844 .017	**.905 .016**

proposed support-guided formulation: stronger latent-space separation does not necessarily imply stronger evidential uncertainty separation. Because KNN neighborhood structure is also used to estimate support during training, we interpret the post-hoc KNN and Mahalanobis results as diagnostics of representation geometry rather than independent evidence of uncertainty reliability. Complete KNN analyses for additional neighborhood sizes and runs are provided in the appendix. Table 5 compares support-only scoring and post-hoc fusion with the jointly support-conditioned model. Support-only scoring and post-hoc fusion demonstrate that latent support contains useful information for OOD discrimination, although their improvements over conventional DER are not consistent across datasets. The jointly support-conditioned model achieves higher mean AUROC than these variants across ApolloScape, SUN RGB-D, and Cityscapes, indicating that the gain is not explained solely by using support as an auxiliary OOD score or post-hoc correction.

3.7. Ablation Study

The default configuration is $M = 512$, $K = 50$, $R = 1000$, $\tau = 8$, and $S_{\min} = 0.05$. These values were fixed as practical design choices rather than selected from the ablation: M and K balance representation coverage with neighbor-search cost, while R balances tracking of the evolving latent manifold with bank-refresh overhead. The remaining parameters control support smoothness and prevent numerical collapse. We then vary each parameter to assess sensitivity around this default configuration. Table 6 reports the sensitivity of Collapse-β OOD performance to each parameter over three runs.

4. Discussion

The experiments indicate that latent representation support provides useful information for evidential uncertainty beyond the predictive head alone. Although support-only and post-hoc DER+support baselines show mixed improvements over conventional DER, the jointly support-conditioned model achieves higher mean AUROC than these variants across SUN RGB-D, ApolloScape, and Cityscapes. This suggests that the gain is not explained solely by using support as an auxiliary OOD score. Instead, conditioning virtual evidence during training allows representation geometry to influence uncertainty generation directly. Because kNN structure is used to estimate support

Table 6. Ablation study of OOD detection performance using Collapse-$\beta = -\log(\beta)$. Results report the mean (μ) and sample standard deviation (σ) of AUROC and FPR95 across three independent runs. M denotes memory-bank size, K the number of nearest neighbors, R the bank-refresh interval, τ the kernel temperature, and $S_{\min}$ the support floor. The default configuration is $M = 512$, $K = 50$, $R = 1000$, $\tau = 8$, and $S_{\min} = 0.05$.

Setting	Value	ApolloScape				Cityscapes			
		AUROC↑		FPR95↓		AUROC↑		FPR95↓	
		μ	σ	μ	σ	μ	σ	μ	σ
M	128	**.895**	**.006**	**.310**	**.034**	.914	.059	.201	.111
	2048	.772	.237	.546	.397	.799	.261	.466	.477
K	5	.844	.088	.508	.178	.885	.081	.331	.224
	100	.647	.239	.584	.270	.764	.124	.472	.132
	300	.794	.237	.418	.274	.890	.147	.223	.244
K,M=100,1024		.827	.125	.372	.085	.755	.151	.440	.067
R	100	.863	.056	.470	.175	.863	.085	.370	.203
	5000	.625	.203	.572	.235	.703	.079	.520	.062
$S_{\min}$	0	.816	.197	.393	.228	.879	.144	.270	.259
	0.02	.771	.205	.509	.251	.847	.156	.318	.249
	0.5	.803	.043	.518	.102	.777	.050	.523	.095
τ	1	.804	.199	.400	.202	.824	.176	.306	.155
	3	.779	.126	.581	.184	.850	.138	.374	.279
	12	.739	.161	.560	.214	.865	.061	.324	.118
Default		.870	.085	.384	.206	**.943**	**.039**	**.138**	**.085**

during training, improved post-hoc kNN separation is partly expected by construction. We therefore interpret kNN and Mahalanobis results as diagnostics of representation geometry rather than primary evidence for OOD performance; our main conclusions are based on evidential and collapse-based uncertainty signals. Progressive training allows the latent representation manifold to organize before support-guided evidential optimization. Compared with direct training from scratch, it improves mean uncertainty-based OOD discrimination across ApolloScape, Cityscapes, and SUN RGB-D. However, the improvement is more pronounced on far-OOD Cityscapes in latent-space KNN analysis, while scratch training performs slightly better on ApolloScape. These results suggest that progressive training benefits uncertainty estimation, although it does not consistently reduce run-to-run variability or improve latent-space separation. Nevertheless, optimization remains sensitive to representation dynamics, as demonstrated by weak and failed runs. Future work could investigate more robust training strategies and stability-aware regularization.

Code Availability. The implementation and supplementary materials are available at https://github.com/fsiar/RepAware_EDL.

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